



**COMSATS University Islamabad,
Sahiwal Campus.**

Auto Transport and Dispatch System (Android Application)

A Project Presented To

**COMSATS University, Islamabad
Sahiwal Campus**

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Software Engineering (2020-2024)

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CERTIFICATE OF APPROVAL

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Executive summary

Auto Transport and Dispatch System revolutionizes the industry by offering a comprehensive, user-friendly, and efficient solution for managing auto shipments. This cutting-edge application optimizes every stage of the shipment process, enhancing operational efficiency, status updating, and customer satisfaction. The functional requirements of the Auto Shipment Management System are the cornerstone of its capabilities, defining the specific behaviors and functions that the system must perform to meet user needs. These requirements encompass essential features such as user authentication, shipment creation, automatic price generator, and real-time status updates. Users, including drivers, admin, and customers, can interact with the system through an intuitive interface, creating and managing shipments seamlessly. The system ensures an efficient automatic price generator.

The app's development will follow an incremental model, allowing for the implementation of the features in stages. The development process will be guided by the functional requirements of the app, user authentication, shipment creation, real-time tracking, route optimization, and communication tools. Users, including drivers, dispatchers, and customers, can interact with the system through an intuitive interface, creating and managing shipments seamlessly.

The app's testing and evaluation will involve unit, integration, and system testing to ensure that all functionalities are working as intended. The app will also undergo user acceptance testing to improve the app's usability.

The auto-shipment system revolutionizes the logistics landscape, offering a powerful, efficient, and user-friendly platform. By seamlessly integrating an automatic price generator, status update, and automated documentation, this application transforms auto shipment management into a streamlined, transparent, and customer-centric process, setting new benchmarks in the industry.

Acknowledgment

All praise is to almighty Allah who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “**Mr. Muhammad Umar**”. Without their supervision, advice, and valuable guidance, the completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Muhammad Bilal

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Abbreviations

OOP	Object Oriented Programming
SDLC	Software Development Life Cycle
UC	Use Case
API	Application Programming Interface
GUI	Graphical User Interface
FYP	Final Year Project
UI	User Interface

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Chapter 1

Introduction

1 Introduction

In today's fast-moving world, getting vehicles from one place to another efficiently is crucial for businesses and individuals. Imagine a world where sending a car, a truckload of goods, or even a single package could be as simple as a few taps on your smartphone. That's exactly what the vehicle shipment system offers. It's like having a personal logistics manager in your pocket, simplifying the way vehicles are transported.

With this user-friendly app, you can effortlessly create and manage shipments. Whether you're a business sending out multiple vehicles or a customer needing a car transported, the process is as easy as entering your details and destination. The app not only manages the whole shipment process but also provides a driver with a separate console to pick loads. You can see exactly where your shipment is and what its status is.

In essence, this android application transforms the hassle of vehicle shipment into a hassle-free experience. It's about simplifying logistics, saving time, and ensuring that your precious cargo, whether it's a car, a truck, or any goods, reaches its destination swiftly and securely. Welcome to the future of effortless transportation right at your fingertips.

1.1 Project Brief

Our vehicle shipment system is a user-friendly android application designed to simplify the process of shipping vehicles efficiently and reliably. With a straightforward interface, the app allows users to easily create, track, and manage vehicle shipments from their smartphones.

The Vehicle Transport Management System is designed to simplify the process of transporting vehicles by creating a user-friendly platform. Customers enter personal details, vehicle specifications, and pickup/delivery preferences, enabling the system to generate an instant quote based on factors such as distance, vehicle type, and selected services. Once a customer confirms and pays for the shipment, the system processes the details and provides a confirmation with scheduled pickup and delivery times. Load information is then presented on the driver console, allowing drivers to accept or reject loads based on their availability. When a driver accepts a load, an appeal is sent to the admin for review. The admin evaluates factors such as driver availability and proximity before approving or rejecting the acceptance, with notifications sent to the driver accordingly. This streamlined process enhances overall efficiency, transparency, and communication in the vehicle transport workflow, minimizing errors through automation.

The primary aim is to streamline vehicle shipment operations. By integrating advanced route optimization algorithms and real-time tracking, the system aims to reduce transit times, minimize fuel consumption, and enhance overall operational efficiency.

The system aims to provide user-friendly interface for customers to input personal details, vehicle specifications, and pickup/delivery preferences. Utilizing factors such as distance, vehicle type, and selected services, the system generates instant and transparent quotes, providing customers with accurate pricing information. Customers receive prompt confirmations with scheduled pickup and delivery times upon confirming and paying for the shipment, streamlining the booking process.

The system aims to facilitate seamless communication between drivers, dispatchers, and customers. Instant messaging and automated notifications ensure that everyone involved is well-informed about shipment updates, reducing communication gaps and enhancing collaboration.

1.2 Relevance to the course module

The main purpose of the project is to provide ease and save time for people having android phone with the help of the android application. So, it is related to the field of software engineering. We have concepts of OOP, software engineering, SDLC, etc. in the development of this android application.

1.3 Background

In a world where moving vehicles from one place to another is common, there's a need for a simple and efficient way to manage these shipments. Traditional methods often involve paperwork, phone calls, and guesswork about vehicle locations. To solve these challenges, our auto shipment system android application was created. It's like a digital assistant for shipping vehicles. This app uses smart technology for automatic price generation, update current status, and keep everyone involved in the shipment process connected. By eliminating the confusion and enhancing communication, this application makes vehicle shipments smoother, faster, and much more reliable for everyone involved. Our system eliminates the confusion by providing driver with separate console in which he gets all loads with complete details and he can generate appeal for picking load.

1.4 Literature review

We have studied systems that are precisely not like these, but some of the modules are implemented in these apps:

- ShipItNow[1]
- QuickLoad Logistics[2]

1.4.1 Analysis review from literature review

Table 1.1:Literature Review

Application Name	Weakness	Proposed Project Solution
ShipItNow	• ShipItNow have manual load assignment, and admin approval delays. And lack of transparent pricing and hidden fees.	• Our application ensures transparent pricing with instant quotes. Our proposed system automates load assignment to drivers
QuickLoad Logistics	• QuickLoad Logistics lacks automated documentation capabilities, relying heavily on manual paperwork for record-keeping. • This manual process is prone to errors and can be time-consuming for administrative tasks.	• Our application automates the documentation process, eliminating paperwork and reducing the chance of errors. • It maintains digital records of shipments, ensuring accuracy and compliance with regulations, saving time for administrative tasks and enhancing overall efficiency.

1.5 Methodology and software life cycle

1.5.1 Design methodology

We will use “Incremental design methodology”. In the incremental model, firstly all the requirements are gathered. After gathering all the requirements and analysis we further move to the next phase which is making the design of a project and then the implementation,

testing, and maintenance phases. As all the next phases of this model are dependent on the previous phase.

In our system we done the development in multiple iterations such as firstly we work on authentication when the whole process of authentication is completed we move to next phase and this iterative development facilitate us to go back and make changes in any previous iteration.

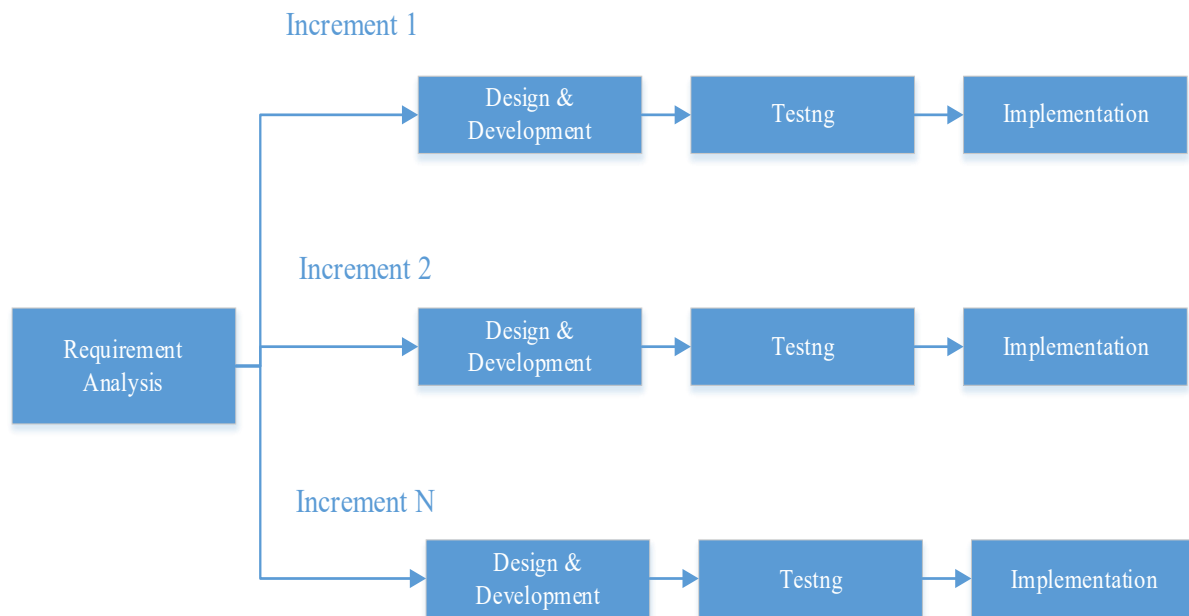


Figure 1.1: Incremental Model

1.5.2 The rationale behind the selected methodology

The incremental software development methodology is an iterative approach to software development that involves breaking down a large project into smaller, more manageable components or increments.

For the Auto Transport and Dispatch System project, the incremental software development methodology was selected for several reasons. Firstly, the project involves developing an application with various modules, each with specific requirements and functionalities. By breaking down the project into smaller increments, the development team can focus on one module at a time, ensuring that the whole authentication process is developed to the highest standard before moving on to the next one.

Secondly, the incremental approach allows us to focus on shipment creation as a next iteration and after completing authentication we move to next this helps to regulate the process flow of applications development.

Each increment can be tested and released independently, allowing for early delivery of working software and faster time-to-market. Overall, the incremental software development methodology was selected for the Auto Transport and Dispatch System project due to its flexibility, adaptability, and ability to deliver a high-quality and reliable product promptly.

Chapter 2

Problem definition

2 Problem definition

2.1 Problem statement

The auto shipment industry faces numerous challenges that hinder its efficiency and customer satisfaction. Existing processes are often plagued by inefficient price generation methodology, customer face lots of problem because many companies tries to contact him at the same time which creates lots of hustle and confusion, no proper source for assigning loads to driver and cumbersome paperwork for documentation. These issues lead to longer delivery times, increased operational costs, customer dissatisfaction due to lack of transparency, and potential errors in compliance-related paperwork. Additionally, the absence of a centralized and user-friendly platform exacerbates these challenges, making it difficult for drivers, admin, and customers to collaborate seamlessly.

Considering these challenges, there is a pressing need for a sophisticated Auto Shipment Android Application. This application must address the inefficiencies pricing techniques, automate documentation processes, and offer an intuitive interface for all users involved. By doing so, it will not only optimize operational processes within the auto shipment industry but also enhance transparency, communication, and overall customer experience. The development of such an application is crucial to revolutionizing the auto shipment sector, ensuring timely deliveries, reducing costs, and fostering trust and satisfaction among businesses and consumers alike.

2.2 Deliverables and development requirements

2.2.1 Deliverables:

- Intuitive and easy-to-navigate design for drivers, admin, and customers.
- User authentication and authorization system for secure access control.
- User manuals and documentation for end-users and administrators.
- Shipment creation and editing features.
- Database for storing shipment data, user profiles, and historical records.
- APIs for communication between the Android app and the backend.

2.2.2 Development Requirements:

1. The app should be developed for the android platform.
2. The app should be user-friendly and intuitive, with a simple and clean interface design that allows users to navigate easily between modules.

3. The app should include the following modules:
 - Shipment management module to develop features for creating, editing, and canceling shipments, allowing users to input essential details such as vehicle type, destination, and delivery time.
 - Implement a secure user authentication system for drivers, admin, and customers, ensuring data privacy and access control.
 - Provide driver with separate console to avoid all confusions.
 - Implement automated notifications and alerts to inform users about shipment status changes and important updates.
 - Price generation module ensures transparency and fairness in pricing, providing both the business and customers with clear insights into the expenses involved in vehicle shipments.
4. The app should allow users to customize their reminders, and notifications according to their preferences.
5. The app should have a reliable and secure database that stores user data and app settings.
6. The app should have a testing phase to identify and fix bugs, as well as optimize the app's performance and speed.
7. The app should comply with the relevant privacy and security regulations and standards.

2.3 Current system

The current systems frequently lack the seamless integration of automated documentation and dynamic pricing, leading to operational bottlenecks and customer dissatisfaction. Unlike these systems, our solution offers automates load assignment to drivers, automates the admin approval process, providing a quick and reliable mechanism for reviewing and approving load assignments, reducing manual intervention and potential delays, automated documentation, and transparent pricing. [3]

Chapter 3

Requirement Analysis

3 Requirement analysis

3.1 Use case Diagram

In this use case diagram, we are showing how a user can find the timetable. There are three types of users: Customer, Admin, and Driver.

3.1.1 Use Case 1 Diagram of Book Shipment

Following diagram show how customer complete shipping process by interacting with system.

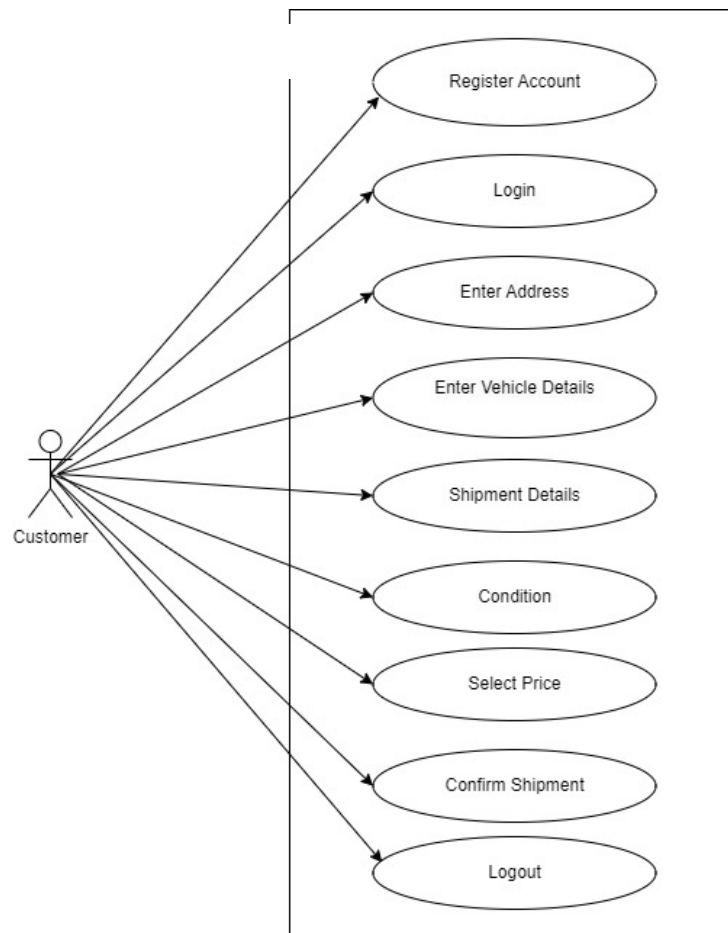


Figure 3.1: User Side use case

3.1.1.1 Use case 1 Description

The table below indicates a comprehensive use case template filled in with an example drawn from use case 1.

These are the use cases for the description:

Table 3.1 :Book Shipment Description

Use Case ID	01
Use Case Name	Book Shipment
Actors	Customer
Description	This use case enables the customer to create a new shipment booking. Users can enter their address, vehicle details, and schedule a shipment for transport.
Trigger	User selects the "Book Shipment" option in the application.
Preconditions	User is logged into their account.
Post conditions	Shipment is scheduled for pickup, and the user receives a confirmation.
Normal Flow	• User selects the "Book Shipment" option in the application. • User enters their pickup and delivery addresses. • User provides vehicle details (type, make, model, etc.). • User selects preferred pickup date and time. • User confirms the booking details. • System calculates the estimated cost of the shipment based on entered information. • System processes the payment, generates a confirmation, and schedules the shipment for pickup.
Alternative Flows	• User can edit the entered information at any step before confirming the booking. • System displays an error message, allowing the user to update payment information or choose an alternative payment method.
Exceptions	• If the user enters an invalid pickup or delivery addresses, the system displays an error message. • If the user enters invalid vehicle details, the system prompts for valid information.
Assumptions	• User has a stable internet connection. • User has valid and accurate address and vehicle details.

	• Payment processing services are available and functional.
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3.1.2 Use case 2 Diagram of Confirm Delivery

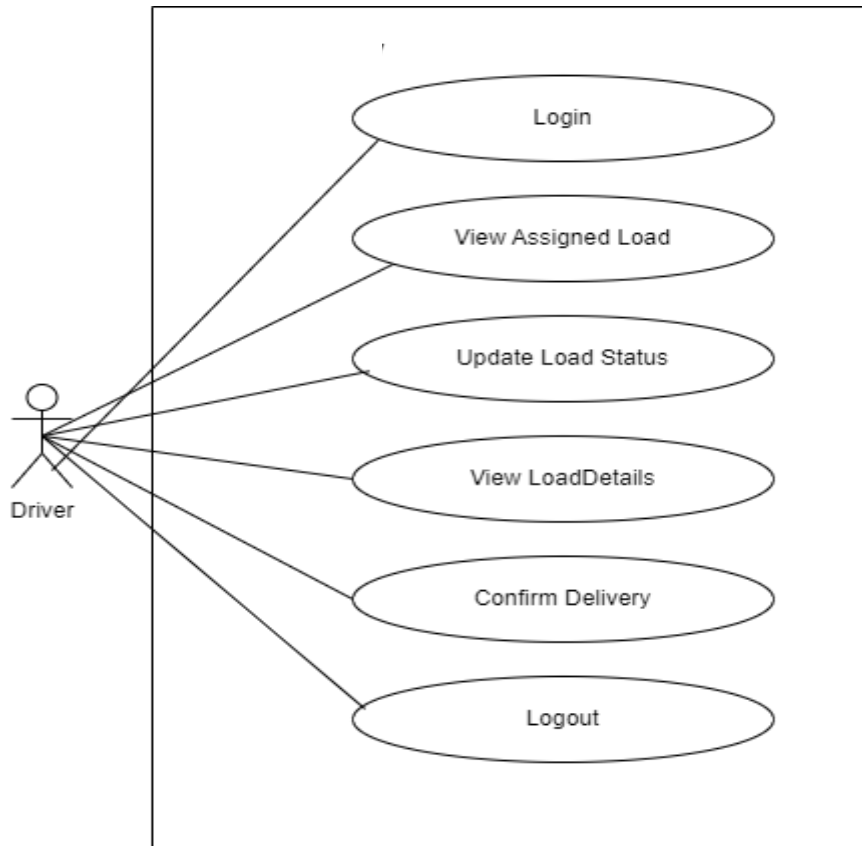


Figure 3.2: Confirm Delivery

3.1.2.1 Use case 2 description

The table below indicates a comprehensive use case template filled in with an example drawn from use case 2.

Table 3.2: Description of Confirm Delivery

Use Case ID	02
Use Case Name	Confirm Delivery
Actors	Driver
Description	This use case represents the interactions of a driver within the application. The driver can log in, view assigned loads, update load statuses, view load details, and confirm deliveries.
Trigger	The use case is triggered when the driver successfully logs into the mobile application.

Pre-condition	- The driver must have a valid account and must be logged into the application. - There must be assigned loads in the system for the driver to interact with.
Post conditions	- Load statuses are updated based on the driver's actions. - Load details are confirmed and updated in the system. - Confirmation notifications are sent to the users associated with the shipments.
Normal Flow	Login: - Driver enters valid credentials. - System authenticates the driver. - System redirects the driver to the dashboard. View Assigned Load: - The driver navigates to the assigned loads section. - The system displays a list of assigned shipments to the driver. Update Load Status: - The driver selects a load from the list. - The driver updates the load status. - System records the status update.
Alternative Flows	- If the driver enters invalid credentials during login. - System displays an error message, and the use case ends.
Exceptions	- If the driver loses internet connectivity during any interaction. - System displays a message indicating a loss of connection.
Assumptions	- Drivers have access to a stable internet connection during interactions. - Load assignments are made correctly and are available in the system for the driver.

3.2 Use case 3 of Admin Management

This figure shows interaction of admin with system.

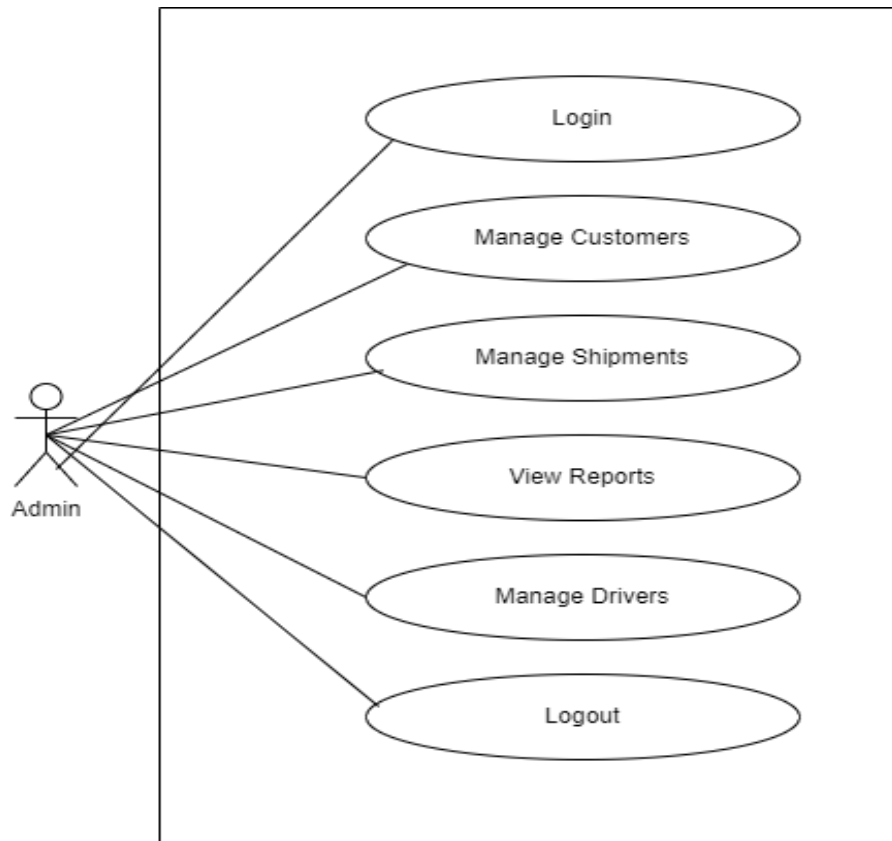


Figure 3.3 :Use case of admin management

3.2.1 Use case description 3

The table below indicates a comprehensive use case template filled in with an example drawn from use case 3.

Table 3.3: Description of admin management

Use Case ID	03
Use Case Name	Admin Management
Actors	Admin
Description	This use case represents the interactions of an Admin within the Auto Shipment Management System. The admin can log in, manage customer accounts, manage shipments, view reports, and manage driver information.
Trigger	The use case is triggered when the admin successfully logs into the Auto Shipment Management System.
Preconditions	- The admin must have valid login credentials. - There must be existing customer accounts, shipments, and driver profiles in the system.

Post conditions	• Customer and driver information is updated as per admin's actions. • Shipments and driver assignments are managed and updated in the system.
Normal Flow	1. Login: • Admin enters valid credentials. • System authenticates the Admin. • System redirects the admin to the admin dashboard. 2. Manage Customer: • Admin navigates to the customer management section. • Admin can add, modify, or delete customer accounts. • System updates customer information accordingly. 3. Manage Shipment: • Admin navigates to the shipment management section. • Admin can view, modify, or cancel existing shipments. • System updates shipment details based on Admin's actions.
Alternative Flows	• If the Admin enters invalid credentials during login. • System displays an error message, and the use case ends.
Exceptions	• If the system experiences a database connection failure during any interaction. • The system displays an error message indicating a technical issue.
Assumptions	Admin has sufficient privileges to manage customer accounts, shipments, and driver profiles.

3.3 Functional requirements

This includes functional requirements of system that depicts functionality of system. This system will have the following modules:

3.3.1 Login Module:

- Secure user login and authentication processes for drivers, dispatchers, and customers, ensuring data privacy and access control.
- Users' login the system by inserting their username and password.
- They can reset their password.
- Users create their new account.
- They will be capable of changing their passwords.

3.3.2 Shipment Management:

- Shipment creation
- Editing
- Cancellation functionalities
- Allowing users to input shipment details, modify existing shipments, and cancel orders when necessary.

3.3.3 Customer Module

- This module is made for customers who want to transport their vehicles.
- This module will take the complete information from them.
- After this, it generates a price for them if the customer accepts the price, then it adds his load to the shipment box.
- The shipment box will contain tracking, driver details and status of his load.
- The customer can also chat with the driver.
- Step-by-step shipment for a vehicle.

3.3.4 Price Generator

- The Price Generator module is a crucial component that calculates the cost of vehicle shipments based on various factors such as distance, vehicle type, and transportation options.
- Price Generator module considers the pickup and delivery locations, vehicle specifications, and customer preferences to generate an accurate and competitive price quote.
- By automating the price calculation process, the Price Generator module eliminates the need for manual price negotiations and provides customers with instant, fair, and reliable pricing information.

3.3.5 Driver Module

- Display a list of available loads on a load board.
- Allow the driver to select a load from the load board.
- Implement a mechanism for the driver to express interest or select a particular load.
- Enable the driver to send an appeal or request for load approval to the admin.
- Generate an agreement and confirmation letter if the admin approves the load request.

3.3.6 Admin Module

- The admin module enables administrators to manage user accounts, including creating new accounts, modifying user information, and handling account-related issues. This goal ensures proper user management and access control within the system.
- Order and Shipment Management: Administrators can monitor and manage the entire lifecycle of vehicle shipments through the admin module.
- This includes tracking the status of orders, managing pickup and delivery schedules, resolving any shipment-related issues, and generating reports for analysis and decision-making.

3.3.7 Logout Module

- All users must use the logout to exit from the application.

3.4 Non-functional requirements

Non-functional requirements are the requirements that specify criteria that can be used to judge the operation of a system. Those constraints under which the system will be operated are called non-functional requirements. For example, language run time environment, operating environment performance requirements, usability requirements, etc. These are all those requirements that do not belong to functional requirements but affect the system overall. We can say some extra conditions and requirements that are not included in the use cases. These are usually called non-functional requirements; some of these are given below.

3.4.1 Usability

The app must have a simple and intuitive user interface that is easy to use for both the customer and driver. It should also be visually appealing and customizable.

3.4.2 Performance

The app should be fast and responsive, with quick load times and minimal lag. It should also be optimized for different screen sizes and device specifications.

3.4.3 Reliability

The app should be reliable, with minimal downtime and errors. It should be able to handle many users simultaneously without crashing or slowing down.

3.4.4 Compatibility

The app should be compatible with the latest version of android and should work seamlessly across different devices and operating systems.

3.4.5 Availability

This app will be readily available to all its users all the time.

3.4.6 System requirements

These requirements consist of the hardware and software components of a computer system that are required to develop and install to use the software efficiently.

3.4.7 Hardware requirements

- Minimum 2 GB RAM to ensure seamless multitasking and application responsiveness.
- Stable 3G, 4G, or Wi-Fi connection for data communication and real-time updates.

3.4.8 Software requirement

- The Internet is necessary to run this application.
- Pre-installed web browser for external links and web-based interactions within the application.

Chapter 4

Design and Architecture

4 Design and architecture

4.1 System architecture

We will discuss what the existing methodologies are and which one we have chosen for the implementation of this project effectively, also we will discuss the advantages of the adopted methodology.

4.2 Data representation

4.2.1 Sequence diagram

The sequence diagram is a visual representation of the interactions between the different objects and components of the system. It shows the flow of messages between the objects and how they collaborate to achieve specific tasks in the system. The sequence diagram captures the dynamic behavior of the system and is useful in identifying potential issues and optimizations in the system's design. [4]

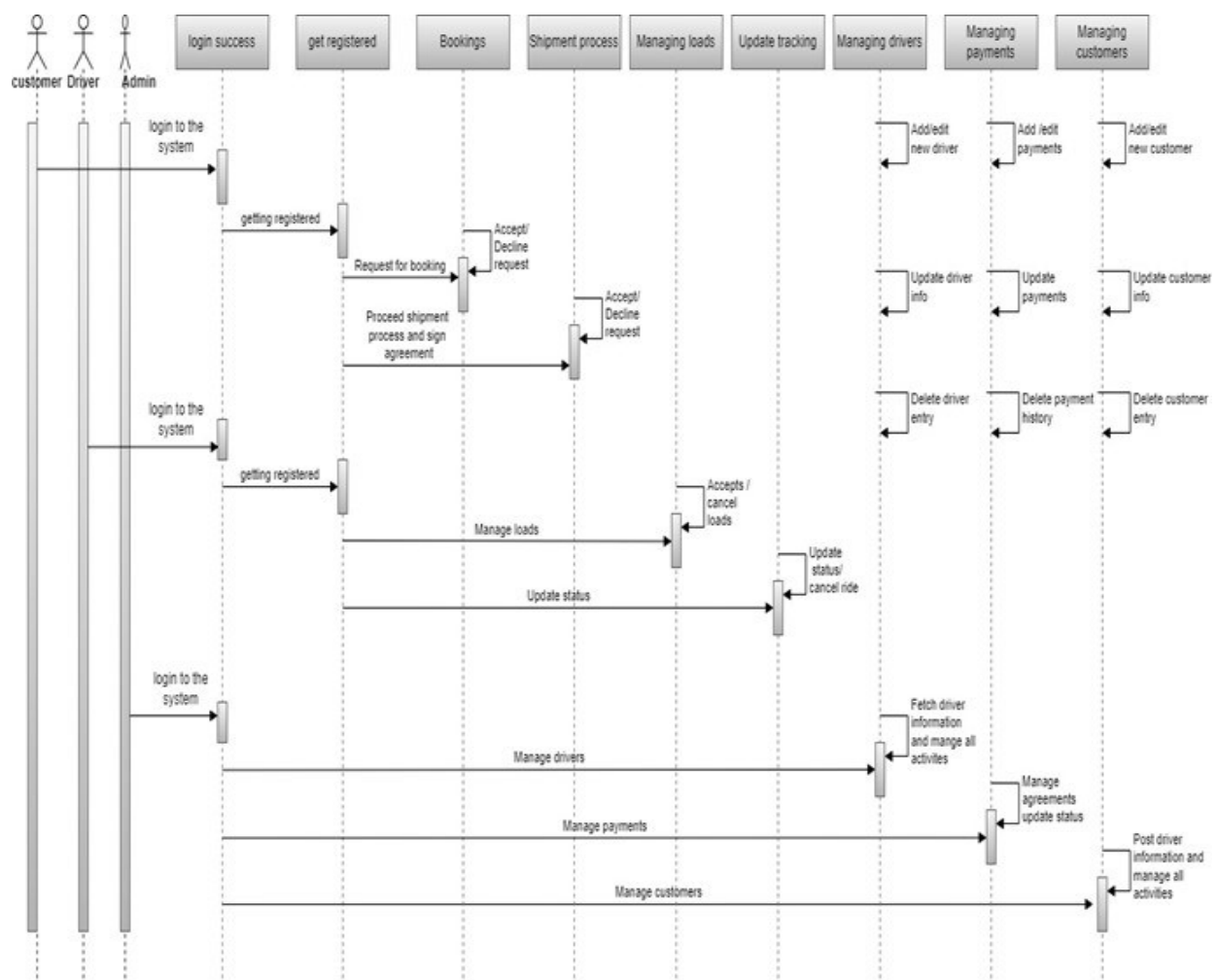


Figure 4.1 :Sequence diagram

4.2.2 Class diagram

The class diagram is a graphical representation of the system's classes, their attributes, and the relationships between them. It depicts the static structure of the system and shows how the different classes in the system are related to each other.

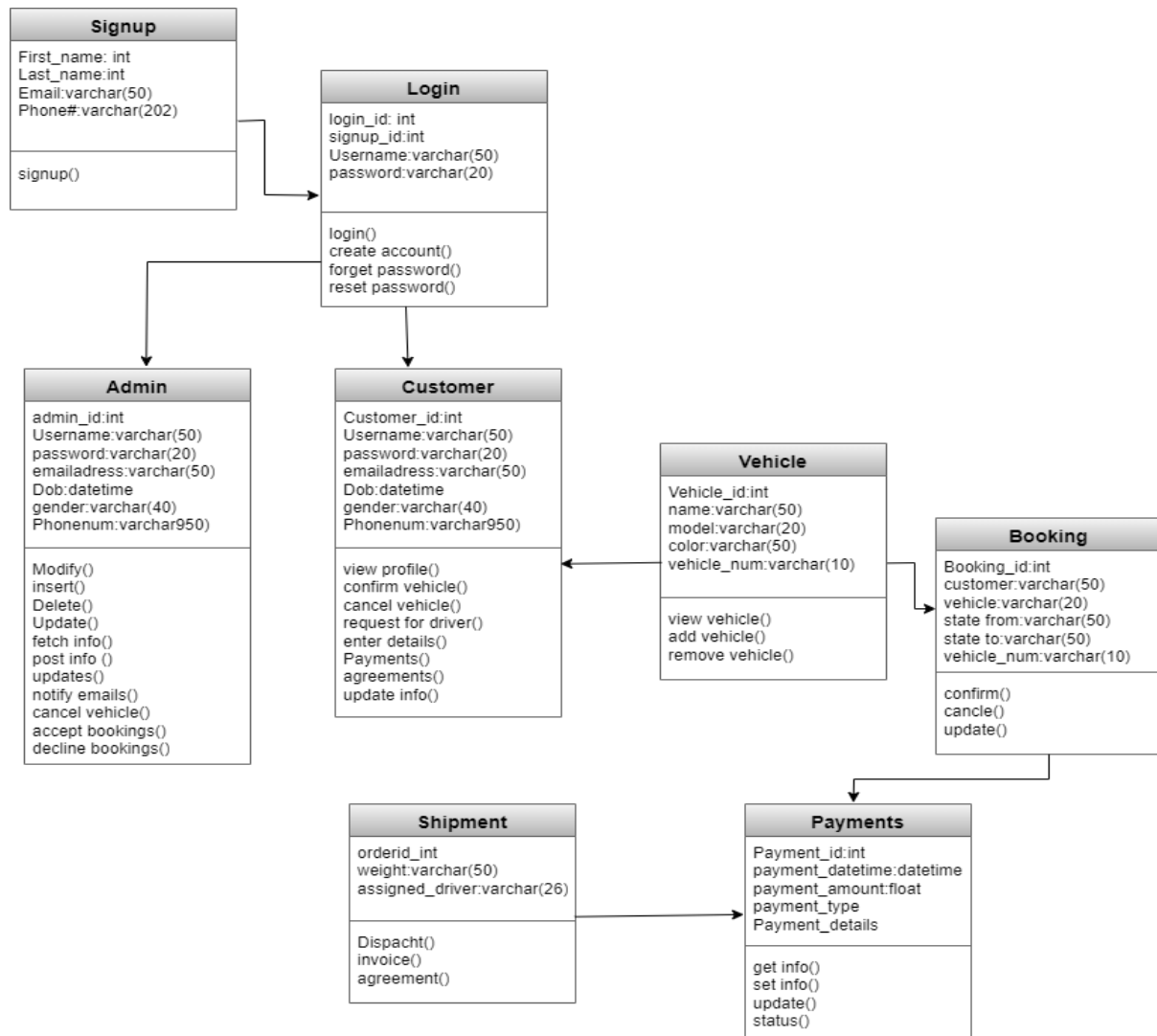


Figure 4.2 :Class diagram of Auto Transport and Dispatch System

4.2.3 Process flow representation

The process begins when a customer creates a new shipment request using the mobile app. The system then optimizes the route, ensuring the vehicle reaches its destination efficiently. Simultaneously, the customer can track the shipment in real-time through the app. The driver, guided by the optimized route, picks up the vehicle and proceeds to the destination. During the journey, the system provides constant updates to both the customer and the dispatch team, ensuring everyone is aware of the vehicle's progress. Once the vehicle reaches the destination, the system generates digital paperwork and receipts, completing the transaction.

The entire process, from creating the request to successful delivery, is seamless and transparent, enhancing efficiency and customer satisfaction.[5]

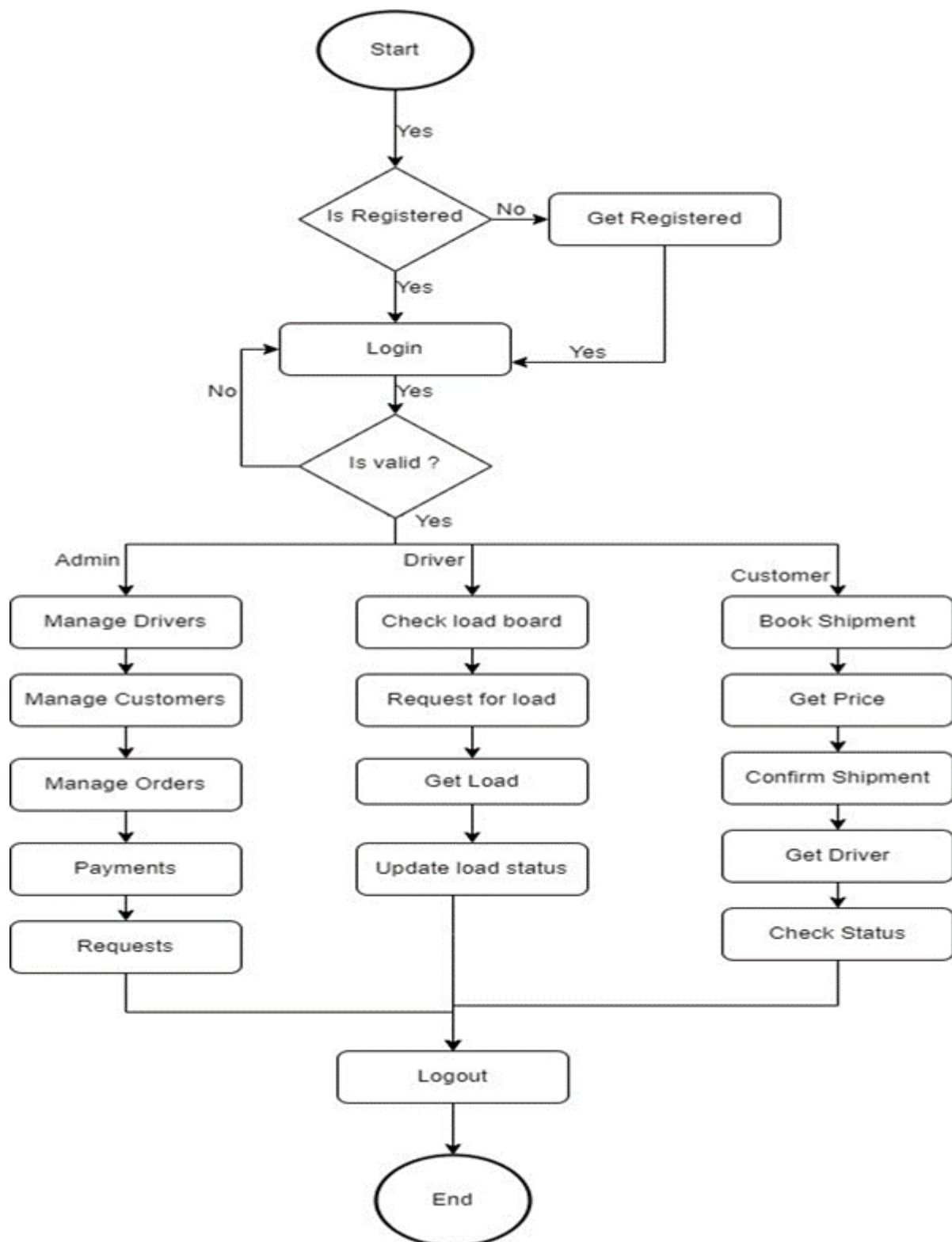


Figure 4.3 :Process flow of Auto Transport and Dispatch System

4.3 Design models

4.3.1 Incremental model

The incremental model was used in the development of the Auto Transport and Dispatch System android app. This model allows for the development of the application in increments, with each increment adding new functionality and features to the app. The model was chosen because it enables the development team to quickly produce a working version of the app, which can be tested and refined before the next increment is added.

Using the incremental model also allows for greater flexibility in the development process, as changes can be made to the app as it is being developed, rather than waiting until the entire app is completed. This helps to reduce the risk of errors and omissions in the final product, as any issues can be identified and corrected early in the development process.

Overall, the use of the incremental model has helped to ensure that the Auto Transport and Dispatch System android app is developed efficiently, with a focus on producing a high-quality product that meets the needs of its users.[6]

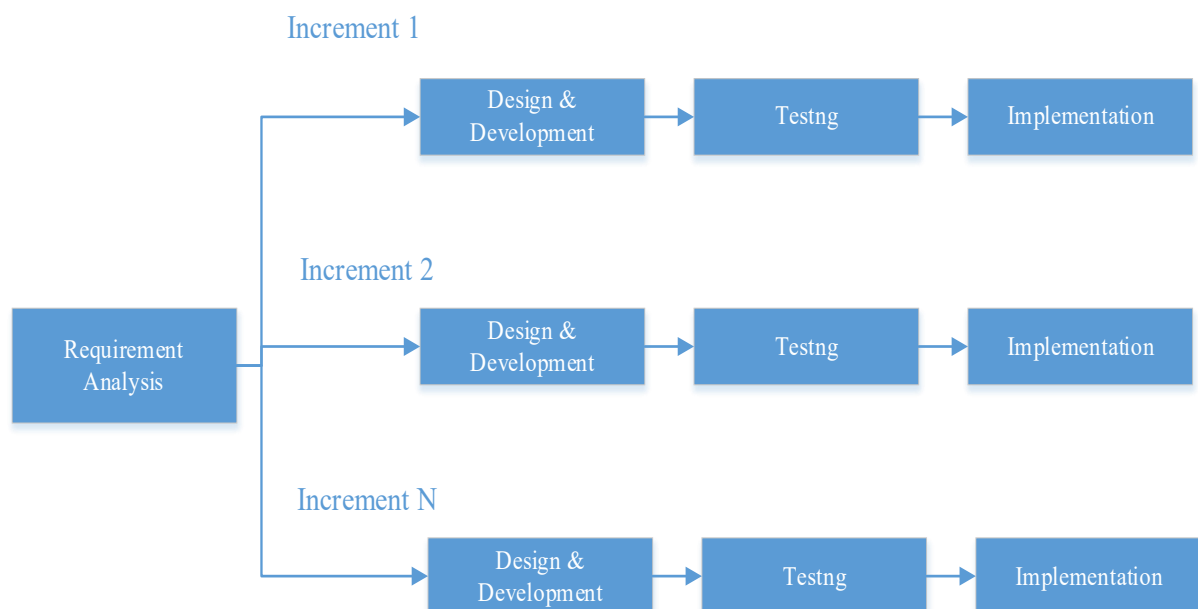


Figure 4.4 :Incremental model

Chapter 5

Implementation

5 Implementation

In this chapter, we'll focus on the implementation of the “**Auto Transport and Dispatch System(android)**” mobile application. Where users can perform many activities on this mobile application.

The most important goal of this phase is to develop the application. The work in this phase should be much more straightforward because of the work done in the planning and design phases. This phase involves changing design specifications into executable programs. When the design is there, developers can have an idea of the looks of the application. All that is needed by developers is to put them in one place to understand.

- The Auto Shipment Management App is developed using Flutter, a robust and versatile framework for building cross-platform applications. Flutter's single codebase approach allows developers to write code once and deploy it on both Android and iOS platforms, ensuring consistency in user experience across devices. Flutter's extensive widget library provides a wide range of customizable UI components, enabling the creation of visually appealing interfaces tailored to the specific needs of the auto shipment management system.
- The database schema is carefully designed to optimize queries, ensuring quick retrieval of information. Additionally, data normalization techniques are applied to eliminate redundancy and improve data integrity. Indexing is utilized strategically to speed up search operations and enhance overall database performance.

5.1 Algorithm

A general algorithm for Auto Transport and Dispatch System is:

5.1.1 User Authentication:

- The user enters their login credentials to access the app.
- The app verifies the credentials against the database and grants access if they are correct.
- If the credentials are incorrect, the user is prompted to re-enter them.

5.1.2 Sign Up Module:

- The user enters their credentials on the sign-up page, typically find fields to input login credentials. These usually include:
 1. Username: Enter your unique username.

2. Email: Enter your email address associated with your account.
3. Password: Create a strong and secure password following the app's password requirements (e.g., length, special characters, etc.).
4. Confirm Password: Re-enter the password to confirm accuracy.

5.1.3 Customer Module

- This module is made for customers who want to transport their vehicles.
- This module will take the complete information from them.
- After this, it generates a price for them if the customer accepts the price, then it adds his load to the shipment box.
- The shipment box will contain tracking, driver details and status of his load.
- The customer can also chat with the driver.
- Step-by-step shipment for a vehicle.

5.1.4 Shipment Management:

- Shipment creation by user.
- Editing in any sort of filled information is entertained.
- Cancellation functionalities.
- Allowing users to input shipment details, modify existing shipments, and cancel orders when necessary.

5.1.5 Price Generator

- The Price Generator module is a crucial component that calculates the cost of vehicle shipments based on various factors such as distance, vehicle type, and transportation options.
- Price Generator module considers the pickup and delivery locations, vehicle specifications, and customer preferences to generate an accurate and competitive price quote.

- By automating the price calculation process, the Price Generator module eliminates the need for manual price negotiations and provides customers with instant, fair, and reliable pricing information.

5.1.6 Driver Module

- Display a list of available loads on a load board.
- Allow the driver to select a load from the load board.
- Implement a mechanism for the driver to express interest or select a particular load.
- Enable the driver to send an appeal or request for load approval to the admin.
- Generate an agreement and confirmation letter if the admin approves the load request.

5.1.7 Admin Module

- The admin module enables administrators to manage user accounts, including creating new accounts, modifying user information, and handling account-related issues. This goal ensures proper user management and access control within the system.
- Shipment management administrators can monitor and manage the entire lifecycle of vehicle shipments through the admin module.
- This includes tracking the status of orders, managing pickup and delivery schedules, resolving any shipment-related issues, and generating reports for analysis and decision-making.

5.2 External API's

Table 5.1 :External APIs

Name of API	Description of API	Purpose of usage
Google Maps API	Google Maps API provides developers with a set of powerful tools to integrate interactive maps, geolocation services, and customized mapping features into their applications and enhancing user	Google Maps API is essential in auto shipment management systems to enable users to monitor the precise location of their shipments. It optimizes routes, calculates distances, and enhances overall

	experience.	logistics efficiency, ensuring timely deliveries and improved customer satisfaction.
Vehicle Data API	The Vehicle Data API is a service that provides comprehensive information about vehicles, including make, model, year, VIN (Vehicle Identification Number), fuel efficiency, and maintenance records.	In auto shipment management systems, Vehicle Data API is crucial for gathering accurate details about the vehicles being shipped.

5.3 User interface

The application's interface will be designed to be straightforward and intuitive, ensuring ease of use for the user. The user will be able to navigate and utilize the application independently without relying on external assistance. Some pages of the user interface are further explained

along
with
images
below:

Figure 5.2 :User interface 1

Figure 5.1 :User interface 2



Figure 5:3: User interface 3

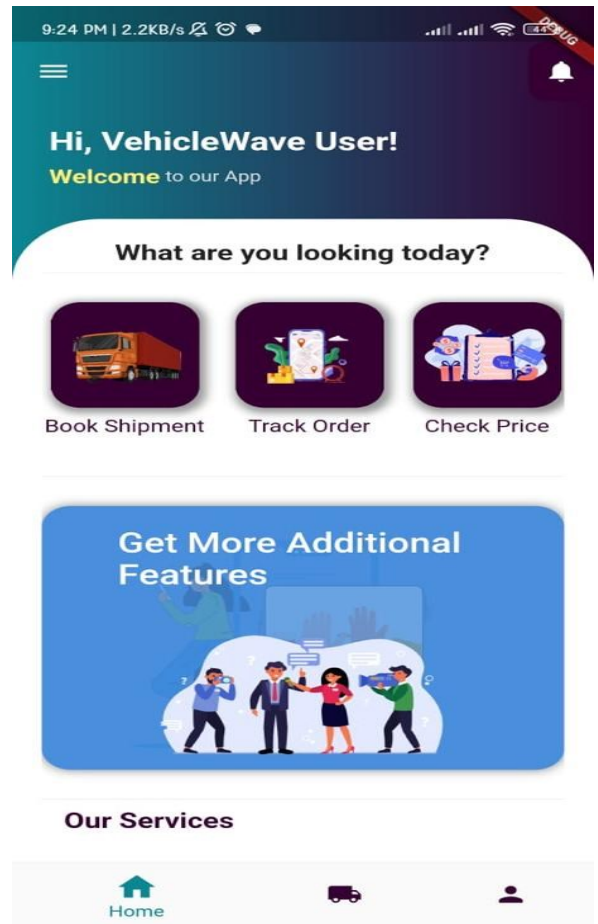


Figure 5:4: User interface 4

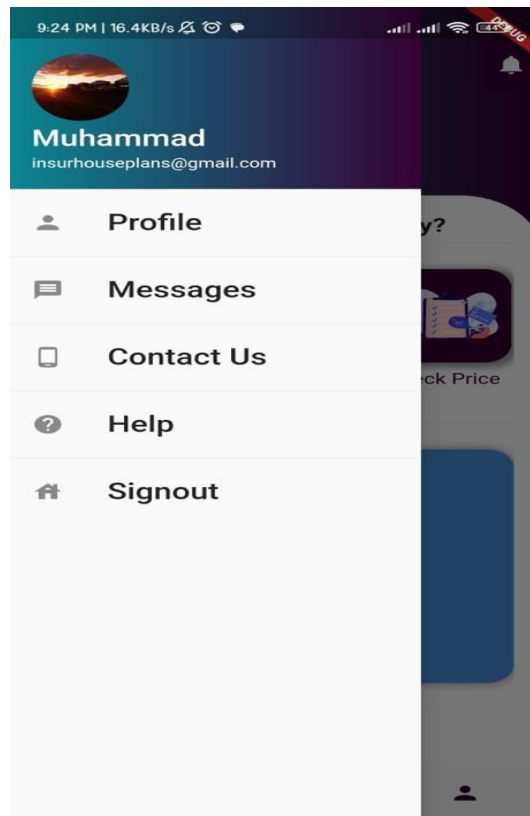


Figure 5.3: User Interface 5

5.4 Screen objects and actions

In the Auto Shipment Android-based application, the screen objects and actions are meticulously designed to ensure a seamless user experience. The main screen features intuitive elements such as buttons for the authentication, booking, and price generation modules. When users access the authentication module, they encounter input fields where they can specify their personal information. Once entered correctly, the screen displays the dashboard of the application. In the Booking module, users navigate through a booking page showcasing various fields for pickup and delivery locations, vehicle details, and preferred dates and times. Authentication actions, such as password verification ensure secure transactions.

For Example, in our application text fields are used for pickup Location that allow users to enter the address from where the vehicle will be picked up. Delivery location enables users to input the address where the vehicle will be delivered. Vehicle details provide space for users to specify vehicle make, model, and other relevant information. Same as date and time pickers are used to pick up dates that allow users to select the date for vehicle pickup.

Dropdown menus are used to select vehicle type users can select the type of vehicle being shipped. Additional services provide options like expedited delivery or insurance, enhancing user choices.

The overall design and layout of the app aim to provide a user-friendly and intuitive experience for users, allowing them to easily navigate the app and access the information they need.

Chapter 6

Testing and Evaluation

6 Testing and evaluation

The testing and evaluation phase is a crucial component of the Auto Transport and Dispatch System android project, ensuring that the developed application meets the desired requirements and is reliable and functional. This chapter outlines the various types of testing that will be conducted during the project, including manual testing, system testing, unit testing, functional testing, and integration testing.[7]

6.1 Manual testing

Manual testing is the process of verifying the functionality of the app manually by the testers. This process is essential to ensure that the app meets the user requirements and performs as expected. The following table shows the test cases and results for manual testing:

Table 6.1: Manual Testing

Test Cases	Expected Result	Actual Result	Pass/Fail
App Installation	Successful	Successful	Pass
App Launch	Successful	Successful	Pass
User Login	Successful	Successful	Pass
User Registration	Successful	Successful	Pass
Search Routes	Accurate	Accurate	Pass
Vehicle Details	Accurate	Accurate	Pass
Add Vehicle	Accurate	Accurate	Pass
Shipment Details	Accurate	Accurate	Pass
Shipment Booking	Successful	Successful	Pass
Payment Processing	Successful	Successful	Pass
Error Handling	Accurate	Accurate	Pass
App Performance and Response	Smooth	Smooth	Pass
App Navigation	Easy	Easy	Pass
App Usability and User Friendliness	High	High	Pass

6.1.1 System testing

System testing is the process of verifying that the application functions as intended in the overall system environment. The following table shows the test cases and results for system testing:

Table 6.2: System testing

Test Cases	Expected Result	Actual Result	Pass/Fail
Compatibility with android Version	Compatible	Compatible	Pass
Compatibility with Device	Compatible	Compatible	Pass
Data Security	Secure	Secure	Pass
Data Backup	Reliable	Reliable	Pass
User Management	Effective	Effective	Pass
Application Response Time	Acceptable	Acceptable	Pass
Application Resource Usage	Optimal	Optimal	Pass

6.1.2 Unit testing

Unit Testing 1: Signup a user

Testing objective: To ensure the user can sign up into the system properly.

Table 6.3 :Unit testing

Test Case	Input	Expected Output	Actual Output	Pass/Fail
1	User will sign up through username email and password	E-mail: Raimafatima95@gmail.com Password: Raima1661	Successfully signup into the system.	Pass
2	User will sign up through username email and password	E-mail: MuhammadBilal786@gmail.com Password: 1234mb	Successfully signup into the system.	Pass

Unit Testing 2: Login into the system.

Testing objective: To ensure the login form works properly.

Table 6.4: Unit Testing of Login

Test Case	Input	Expected Output	Actual Output	Pass/Fail
1	User will login through email and password or direct to Google.	E-mail: raimafatima6262@gmail.com Password: raima@62	Successfully login into the system.	Pass

Unit Testing 3: Booking with invalid Input.

Testing objective: To ensure that system shows an error message.

Table 6.5: Unit Testing of Booking

Test Case	Input	Expected Output	Actual Output	Pass/Fail
1	Valid pickup and delivery details, valid vehicle type, but past date/time for pickup.	Booking should fail due to invalid date/time, and an error message should be displayed.	App displays an error message indicating an invalid pickup date/time. Booking fails.	Pass
2	Valid pickup and delivery details, invalid vehicle type, and preferred date/time.	Booking should fail due to an invalid vehicle type, and an error message should be displayed.	App displays an error message indicating an invalid vehicle type. Booking fails.	Pass

6.2 Integration testing

Integration testing is a critical phase in software development where individual components or modules of a software application are combined and tested as a group to ensure they work together seamlessly.

Table 6.6: Integration Testing

Test Case name	Signup a user
Test Case number	01
Description	The user can sign up into the system with enter his/her basic information like username, email and valid password.

Testing technique used	Manual testing, Interface Testing using the black-box testing technique, Condition Coverage testing.
Precondition	Users should have valid email and password for signup.
Input values	Enter email xyz@gmail.com Enter password 12345
Steps	• Users need to on their internet services. • User must need valid username, email and password on the dashboard to click on the signup button and enter his/her basic information. • The status is signup successfully
Expected output	Sign up successfully
Actual output	Status changed.
Status	Pass

6.3 Functional testing

Functional testing will be conducted to ensure that the application meets the functional requirements specified in the project scope. The testing will cover all modules of the application, including user module, pricing module, admin module, shipment management module and authentication module. The test cases for functional testing are as follows:

Table 6.7: Functional Requirement

Test ID	Test Description	Expected Result
F01	Verify that the user can login into the system	The user can login into the system.
F02	Verify that the user can sign up into the system.	The user can correctly sign up into the system.
F03	Verify that the user can reset the password.	The user can reset the password.
F04	Verify that the user can select pick up and drop off address.	User can select any valid address.
F05	Verify user can view past shipment history.	User should be able to access and view their past shipment details.
F06	Verify user receives notifications for shipment status updates.	User should receive timely notifications about the status of their active shipments.
F07	Verify the accuracy of the estimated shipment cost calculation.	The estimated cost displayed after entering shipment details should match the calculated cost based on distance and vehicle type.

F08	Verify the functionality of adding and managing multiple shipments simultaneously.	User should be able to add and manage multiple shipments concurrently without any conflicts or data overlapping.
F09	Verify the accuracy of the delivery time estimation.	The estimated delivery time provided to the user should be accurate, considering factors such as distance and delivery route.
F10	Verify the secure handling of sensitive user data, such as passwords and payment information.	User data should be encrypted and securely stored, and there should be no unauthorized access to sensitive information.
F11	Verify the functionality of the date and time selection during shipment booking.	User should be able to select any valid future date and time for pickup, ensuring the system accepts the chosen schedule accurately.
F12	Verify user can log out of the application.	User should be successfully logged out, terminating the current session.

Chapter 7

Conclusion and Future Work

7 Conclusion and future work

7.1 Conclusion

The auto shipment system is to simplify and streamline the process of shipping vehicles from one place to another. By providing users with a user-friendly platform, the system aims to enable effortless booking and efficient management of vehicle shipments. It serves businesses and individuals by ensuring secure, timely, and convenient transportation of vehicles, ultimately enhancing customer satisfaction, and optimizing the logistics involved in auto shipments.

By automating tasks and providing a centralized platform, the system reduces manual efforts, minimizes errors, and optimizes the logistics chain. It enables businesses to efficiently manage their fleet, ensures timely deliveries, and enhances customer communication through real-time tracking and notifications. Additionally, it promotes cost-effectiveness by optimizing routes and resources, making the vehicle shipping industry more efficient and reliable for both providers and customers.

In conclusion, the auto shipment android application is well-equipped to meet user expectations and deliver a smooth, efficient, and secure experience in the realm of vehicle logistics. The meticulous testing efforts have validated the application's functionality, ensuring it is ready for deployment, contributing to a seamless user experience, and fulfilling the needs of its users in the auto shipment domain.

7.2 Future work

We plan to add modules to this application in the future. These modules are:

- Tracking Module
- Chat Module

These modules will enhance the functionality of the application, making it more user-friendly and efficient. The Tracking Module significantly improves system efficiency by offering real-time visibility into shipment locations for both users and carriers. The Chat Module enhances efficiency by enabling instant communication between users, carriers, and customer support within the application. Overall adding these modules will provide a more comprehensive and complete user experience, and we are excited to implement them soon.

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